Maintaining state during postbacks

As a reminder, `OnGet` handlers execute for `GET` requests, and `OnPost` handlers execute for `POST` requests. As part of form management, you will often need to initialize data in the `OnGet` handler, such as preparing options for select lists and so on. Once the framework has used that data as part of the HTML generation process and the response has been sent to the browser (step 2 in figure 5.2), any data that was generated on the server is lost. This is because, by default, Razor Pages is stateless. That is, it doesn't maintain that data anywhere. This makes sense because HTTP is a stateless protocol.

If you want to work with the same data in the `OnPost` handler, you must re-initialize it within the `OnPost` handler. Failure to do so may result in runtime exceptions as you try to reference data that doesn't exist because you haven't created it! This most often tends to happen when the form submission fails validation checks and needs to be presented again. It can be easy to overlook the fact that select lists, in particular, often rely on data generated on the server, which needs to be generated each time you display the form.

Clearly, a vital part of form processing is being able to access the submitted values, so you can validate and process them. In this section, I show how they are assigned to the `Request` object, which, as you will remember, is a property of the `HttpContext`, along with the various options available to you for retrieval of these values for `POST` and `GET` requests. While you will access the `Request` object directly in the following examples, this is not the recommended approach when working with form data in a Razor page. However, once you understand the limitations of this approach, the role of model binding—the approach you should use as your default—and the benefits it brings will make a lot more sense.

5.1.2 Accessing values from Request.Form

When a form is submitted using the `POST` method, the form values are processed by the underlying ASP.NET Core framework and assigned, with